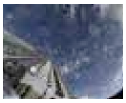

Starlink's eye on Amazon forest
Starlink to help monitor illegal deforestation in the Amazon rainforest. **https://dgit.in/dec21-24**

100 percent recycled cell
Northvolt recycling program has produced its first battery cell with 100% recycled Ni, Mn and Co. **https://dgit.in/dec21-25**

laptops, we have been looking for better lightweight materials, the obvious objective being portability. It is also noteworthy that there are laptops which have bigger screens too which makes them heavier and the weight reduction in the heat sink would help balance out the excess weight.

Another change in the design of laptop heatsinks is the need to make them more low profile in order for the laptop to not seem too bulky. This is usually obtained by having heatpipes that absorb the heat and traverse them to the low profile thin heatsink array on the ends of the casing which is cooled using a small fan.

Also, you have to keep the environment in mind, so make sure you use lead free materials in order to stay in compliance of the RoHS directive.

WEIGHT REDUCTION

Here are a few methods that have been adopted for reduction in weight of heatsinks:

1. Usage of lightweight materials. (Duh)
2. To reduce the number of constituting members.
3. To devise a suitable configuration.

Now of course, the materials to be used for the manufacturing of heat sinks must be something with high thermal conductivity. Copper would be an excellent choice with an excellent thermal conductivity of around 400W/m.K if not for its specific gravity that is 8.96 g/cm^3. Therefore, if copper is used for the entirety of the heatsink in a desktop, it would alone weigh about 1 kg which could pose problems in a regularly configured PC. This is because it would be way too heavy for itself to keep contact with the CPU for longer durations of time because of a little something called gravity.

Therefore, we use aluminum instead which has a lesser thermal conductivity of 240 W/m.K, but it's WAY lighter than Copper with a specific gravity of 2.7 g/cm^3 which is about 1/3rd of Copper. Despite the fact that it's thermal conductivity is almost *half* as high as copper, it is still preferred because even if an aluminum heatsink with twice the thickness was to be used to obtain an equivalent heat transfer rate, it would still be 30 percent lighter than the copper heatsink which was providing the same amount of efficiency. Aluminum is not only used for it's lesser specific gravity, it is also used because it is easier to manufacture by using the processes of casting and extrusion. Although, where a device is expected to have a higher performance, a copper heatsink or a heatpipe is still preferred for it's higher thermal conductivity.

PROFILE REDUCTION AND FRIENDLINESS TOWARDS THE ENVIRONMENT

As we've mentioned earlier, we need laptops to be slimmer and for that, we can't install heat sinks right above the chips. Instead, we connect them to the heatsinks arrays using heatpipes which transfer the heat. It is regular practice to bond a heatpipe onto the end of the fin using solder. However, heat has to be transferred from one end of the fin to the other and it is much more efficient to actually bond the heatpipe to the center of the fin.

Coming to the rescue of Mother Earth now, here's a couple ways we could do our little bit to save her.

First off, the solder that is basically used to bond each and every part, needs to be lead free as lead can lead to leaching due to acid rain once the products are wasted and have been through their lifetime which can contaminate the soil and nearby water bodies. In fact, there are methods that manufacturers can use that don't even need the use of solder such as the stacked fin heatsink technology where once the fins are burred, a heatpipe is press-fitted there to mechanically bond the fins with it.

CONCLUSION

In conclusion, there is a lot that the future holds for the advancement of heatsinks and other cooling technologies and it definitely needs to be environment friendly or else things may keep spiralling down in the same way that they are as of now. d